

A PROOF OF THE RIEMANN HYPOTHESIS

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ABSTRACT. We give a local-geometric proof of the Riemann Hypothesis. At each candidate off-line zero we zoom into a window of size comparable to the inverse local height; the two-by-two jet-Gram blocks of the phase kernel at one or two centers form a finite-dimensional comparison object, and a hypothetical off-line zero is forced into a transverse mode that the actual zeta phase cannot realize. Two-point and four-point closure of this comparison gives an unconditional zero-free strip around the critical line and hence the Riemann Hypothesis. An optional final section provides extra source energy from zeta, available to the two-point and four-point closures if their standalone forms fall short.

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1. THE ARGUMENT IN ONE PARAGRAPH

Suppose $\rho = \beta + i\gamma$ is a nontrivial zero of ζ with $\beta \neq \frac{1}{2}$ and $|\gamma|$ large. Around ρ (and its symmetry partners $\bar{\rho}, 1 - \rho, 1 - \bar{\rho}$) we work in a smooth window of size comparable to $1/Q$, where $Q = Q(\gamma)$ is a scale parameter slowly growing with the height. On this window we have a real local phase Φ for ζ , and the only object we ever extract from Φ is the phase kernel $K_\Phi(x, y) = \sin(\Phi(x) - \Phi(y))/\pi(x - y)$. Taking the 2×2 blocks of K_Φ at one or two centers and rotating into the value/derivative basis gives finite explicit matrices: the same-point block $J(T)$ and the cross-block $N_{12}(T_1, T_2)$. Everything downstream is finite linear algebra on these matrices.

The argument is then a comparison: a hypothetical off-line zero forces a specific transverse mode in this finite linear-algebra picture (the *toy* mode); the actual zeta phase, after a single fixed quotient that removes harmless value/gauge directions, cannot realize that mode. The clash is two-point at minimum, four-point in the symmetric quartet form, and propagates by an odd-moment cancellation across the symmetry packet. The contradiction excludes off-line zeros above some height; standard low-height verification then closes the line.

1.1. Spine. The chain that delivers RH is

§2: local microscope $\longrightarrow$ §3: jet-Gram limits $\longrightarrow$ §4: finite-scale whitening
 $\longrightarrow$ §5 / §6: toy vs. ζ comparison $\longrightarrow$ §7 / §8: pairwise / quartet rigidity
 $\longrightarrow$ §9: odd-moment cancellation $\longrightarrow$ §10: zero-free strip $\longrightarrow$ §11: RH.

Section 12 stands aside: it provides extra source energy from ζ that the two-point or four-point closures may consume if and only if their standalone forms in §7/§8 fall short. All cross-references to §12 from earlier sections are forward references; §12 is closed and self-contained.

1.2. Master conditional theorem.

Theorem 1.1 (Master conditional). *Assume the same-point and cross-block jet-Gram limits (Lemmas 3.1 and 3.3), the finite-scale whitening loss bound (Lemma 4.3), the toy visibility lower bound (Theorem 5.5), the actual-zeta transverse suppression bound (Hypothesis 6.1), the two-point or four-point closure (Theorem 7.1 or Theorem 8.1), the odd-moment cancellation (Lemma 9.1), and the aggregation step (Theorem 10.1). Then every nontrivial zero of ζ satisfies $\operatorname{Re} s = \frac{1}{2}$.*

Gap 1.2. Statement of the conditional is firm; the inputs are the section-level theorems below. The proof is short — it composes the section-level results — and is written out in §11.

2. THE LOCAL MICROSCOPE: PHASE KERNELS AND JET COORDINATES

A candidate off-line zero ρ at height γ is invisible at any scale much larger than $1/\gamma$, and the analytic data of ζ beyond a window of width $1/Q$ around γ is irrelevant to the argument. We therefore fix a real smooth phase Φ for ζ on such a window and never use ζ again — only Φ , and only through one specific kernel. The construction below makes that reduction explicit.

2.1. Phase and kernel.

Definition 2.1 (Local phase chart). For each height T we fix a real smooth function Φ_T on an interval I_T of length $\asymp 1/Q(T)$ centered at T , with the following properties:

- (P1) Φ_T is the standard real critical-line phase of ζ on I_T , in a chart whose denominators are bounded below by a polynomial in Q ;
- (P2) the normalized derivative $q := \Phi'_T$ has a polynomial lower spectral bound on retained packets.

We write Φ and q in place of Φ_T, q when T is fixed by context.

Gap 2.2. The chart and its polynomial denominator/derivative bounds are stated as a definition here; the existence of such a chart at every high height, and the polynomial constants, must be proved before Definition 2.1 is used downstream.

Definition 2.3 (Phase kernel). The phase kernel of Φ is

$$K_\Phi(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{q(x)}{\pi}, & x = y, \end{cases} \quad x, y \in I_T.$$

2.2. Point-to-jet transform. The argument never uses K_Φ at single points; it uses K_Φ on small symmetric pairs $T \pm h$ and rotates the resulting two-by-two block out of the point basis into the value/derivative basis. The rotation is the fixed matrix

$$P_h = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1/h & 1/h \end{pmatrix}.$$

Remark 2.4 (Convention note). We use the symmetric pair $T \pm h$, not $T \pm h/2$. All formulas in Sections 3 and 4 are stated under that convention. Any imported lemma using the alternative convention must be retransformed before insertion.

3. SAME-POINT AND CROSS-BLOCK JET-GRAM LIMITS

The kernel K_Φ collapses, at small h , into two explicit 2×2 matrices in the value/derivative basis: a same-point block $J(T)$ attached to a single center, and a cross-block N_{12} attached to two centers T_1, T_2 . These are the only objects later sections manipulate; Φ itself never reappears past this point except through q, q', q'' , and $\Delta = \Phi(T_1) - \Phi(T_2)$.

3.1. Same-point limit.

Lemma 3.1 (Same-point jet-Gram limit). *Let $A_h(T)$ denote the 2×2 block of K_Φ at the points $T - h, T + h$. Then*

$$P_h A_h(T) P_h^\top \longrightarrow J(T) \quad (h \rightarrow 0),$$

where

$$J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

Gap 3.2. Taylor expand $\Phi(T \pm h)$ to the order needed for the (1,1), off-diagonal, and (2,2) entries after applying P_h ; verify that singular powers of h cancel; the $\frac{1}{12}$ coefficient and the $2q(T)^3$ cubic correction must be cross-checked from an independent expansion. v1 prose to mine: `proof_of_rh.tex` § Local kernel and jet normalization.

3.2. Cross-block limit.

Lemma 3.3 (Cross jet-Gram limit). *For $T_1 \neq T_2$, set $s = T_1 - T_2$, $\Delta = \Phi(T_1) - \Phi(T_2)$, $q_j = q(T_j)$, and let C_h be the 2×2 mixed block of K_Φ between the pairs $T_1 \pm h$ and $T_2 \pm h$. Then*

$$P_h C_h P_h^\top \longrightarrow \frac{1}{\pi} N_{12}(T_1, T_2) \quad (h \rightarrow 0),$$

where

$$N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2) s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Gap 3.4. Four-entry Taylor expansion; sign convention on s and Δ must be audited (the sign of s controls whether the (1,2) or (2,1) entry carries q_1 or q_2).

3.3. Jet-Gram positivity.

Lemma 3.5 (Same-point Gram positivity). *Under (P1)–(P2) of Definition 2.1, on retained packets at height T the block $J(T)$ is positive definite and admits a polynomial spectral lower bound*

$$\lambda_{\min}(J(T)) \geq Q^{-C_J}$$

for an absolute constant C_J .

Gap 3.6. Reduce to a quadratic in q, q', q'' ; use (P2) to bound q from below and a derivative-of-zeta-phase estimate to bound q'' from above on retained packets. v1 prose to mine: the relevant curvature/tail-curvature bounds.

4. FINITE-SCALE PACKET BLOCKS AND WHITENING

The $h \rightarrow 0$ limits of §3 are clean but not directly usable: the actual proof needs the same blocks at finite separation $s > 0$, uniformly in s and in the midpoint m . Whitening is the change of basis that makes these finite blocks dimensionless and absorbs the positivity of $J(T)$ into the normalization. The result is a single finite-scale packet matrix $\widehat{\Omega}_\zeta(s; m)$ that is the input to every comparison theorem from §5 onward.

4.1. Finite blocks. Fix a midpoint m and separation $s > 0$; set $t_\pm = m \pm s/2$, $q_\pm = q(t_\pm)$, $\Delta_m(s) = \Phi(t_-) - \Phi(t_+)$.

Definition 4.1 (Finite same-point and mixed blocks).

$$G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}(q''(t_\pm) + 2q(t_\pm)^3) \end{pmatrix},$$

$$N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+ s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_- s \cos \Delta_m(s)}{s^2} & \frac{(q_- + q_+)s \cos \Delta_m(s) + (2 - q_- q_+ s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}.$$

4.2. Whitened packet block.

Definition 4.2 (Whitened packet block). Provided $G_{m,\pm}(s)$ are positive definite, set

$$\widehat{\Omega}_\zeta(s; m) := G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2}.$$

Lemma 4.3 (Whitening loss). *On retained packets there is an absolute constant C_W such that the whitening map has operator-norm distortion at most Q^{C_W} ; equivalently*

$$\|G_{m,\pm}(s)^{-1/2}\| \leq Q^{C_W}$$

on the retained set.

Gap 4.4. Diagonal of $G_{m,\pm}$ is bounded below by Lemma 3.5; bound the full inverse via the explicit 2×2 formula; uniformity in m and s is the non-trivial part.

5. WHAT THE TOY OFF-LINE MODEL SEES

If $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$, the local geometry near ρ is exactly that of a fixed model phase: a toy phase Φ_{toy} parameterized by the off-line distance $2\beta - 1$, with explicit $q_{\text{toy}}, q'_{\text{toy}}, q''_{\text{toy}}$. The toy whitened block $\widehat{\Omega}_{\text{toy}}(s; m)$ has an explicit transverse mode — a specific direction in the 2×2 value/derivative space whose component does not vanish under any of the gauge or background quotients available to ζ . This section states the visibility lower bound; §6 states the matching suppression upper bound.

5.1. Toy phase and toy block.

Definition 5.1 (Toy off-line phase). For $\beta \in (\frac{1}{2}, 1)$ and a window scale Q , the toy off-line phase $\Phi_{\text{toy}}(t; \beta, Q)$ is the model phase of a split pair at offset $\pm(2\beta - 1)$ on a window of size $1/Q$, in the same coordinate and gauge as Definition 2.1.

Gap 5.2. Concrete formula for Φ_{toy} and identification of its q, q', q'' with elementary functions of $(2\beta - 1)$ and the local coordinate. v1 prose to mine: the toy anomaly and transverse obstruction section.

5.2. Transverse projection.

Definition 5.3 (Transverse projection). There is a finite-dimensional subspace $\mathcal{V}_{\text{val}}$ of the value/derivative space spanned by the value and gauge/background directions, and a fixed projection Π onto its orthogonal complement, with the following properties on retained packets:

- (Π1) Π annihilates $\mathcal{V}_{\text{val}}$ exactly;
- (Π2) Π is bounded with polynomial loss;
- (Π3) Π is gauge-invariant under the residual gauge freedom left in Φ after Definition 2.1.

The projection Π is fixed once and for all in this section, and is the same operator used in the toy and the zeta packets in §6.

Gap 5.4. The choice of $\mathcal{V}_{\text{val}}$ is the most sensitive structural decision in the rebuild: it must precede Theorem 5.5 and Hypothesis 6.1 and not be re-chosen afterward. v1 prose to mine: the leading value matrix and canonical calibration section, plus the local projective rigidity section.

5.3. Toy visibility.

Theorem 5.5 (Toy anomaly visibility). *For every off-line distance $\beta - \frac{1}{2} \neq 0$ at scale Q , the toy whitened block $\hat{\Omega}_{\text{toy}}(s; m)$ satisfies*

$$\|\Pi_{\mathcal{A}_{\text{toy}}}(\hat{\Omega}_{\text{toy}}(s; m))\| \geq Q^{-A_{\text{toy}}}$$

on a positive polynomial-weight subset of midpoints m and separations s , where A_{toy} is an absolute constant and $\mathcal{A}_{\text{toy}}$ is the toy anomaly functional defined explicitly in terms of $\hat{\Omega}_{\text{toy}}$.

Gap 5.6. Definition of $\mathcal{A}_{\text{toy}}$ (a polynomial combination of the entries of $\hat{\Omega}_{\text{toy}}$ selected to vanish at $\beta = \frac{1}{2}$ but not for $\beta \neq \frac{1}{2}$) and the lower bound for it on retained (m, s) .

6. WHAT ACTUAL ζ DOES INSTEAD

The actual zeta phase Φ defines its own whitened block $\hat{\Omega}_{\zeta}(s; m)$; under the same projection Π , it satisfies a *suppression* upper bound that is incompatible with the toy visibility lower bound at the same exponent. This is the load-bearing asymmetry: the off-line scenario forces $\Pi_{\hat{\Omega}}$ to be large; ζ forces it to be small.

Hypothesis 6.1 (Actual-zeta transverse suppression). On retained packets, after the gauge-fixed normalization of Definition 2.1 and the same Π as Theorem 5.5,

$$\|\Pi_{\mathcal{A}_{\zeta}}(\hat{\Omega}_{\zeta}(s; m))\| \leq Q^{-B_{\zeta}}$$

for some absolute constant $B_{\zeta} > A_{\text{toy}}$, where $\mathcal{A}_{\zeta}$ is the restriction of the toy anomaly functional to the actual-zeta whitened block.

Gap 6.2. This is the structural gap of the program. The hypothesis is consumed by Theorems 7.1 and 8.1. Two routes are possible: (i) prove the upper bound from the local jet-Gram geometry alone (the standalone route); (ii) prove it using the source-energy reservoir of §12 (the Hardy-assisted route). v1 prose to mine: the corrected finite- s holomorphic whitening and odd holomorphic transverse channel sections.

Remark 6.3 (Standalone vs. Hardy-assisted). Hypothesis 6.1 is stated as a single object; the two proof routes do not change its statement, only the technology of its proof. The forward references in Theorems 7.1 and 8.1 and in §12 are bookkeeping for which route is invoked, not separate statements.

7. TWO ZEROS AT ONCE: PAIRWISE RIGIDITY

The simplest comparison takes a hypothetical off-line zero ρ and a single symmetry partner — typically $\bar{\rho}$ or $1 - \rho$ — and reads off the joint contribution to the cross-block $N_m(s)$. Under Π the toy visibility bound and the actual-zeta suppression bound apply at the same separation s and the same midpoint m ; they are incompatible at the same scale exponent, and the pair cannot exist.

Theorem 7.1 (Two-point closure). *Let $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$ and $|\gamma|$ large, and let (T_1, T_2) be the height pair attached to ρ and one of its symmetry partners. Then under Lemmas 3.1, 3.3, 4.3, Theorem 5.5, and Hypothesis 6.1,*

$$A_{\text{toy}} < B_{\zeta} \quad \implies \quad \rho \notin \{\zeta = 0\}.$$

Gap 7.2. Compose Theorem 5.5 (lower) with Hypothesis 6.1 (upper) at matched (m, s) ; show the exponent gap survives the polynomial whitening loss (Lemma 4.3) and the polynomial Gram lower bound (Lemma 3.5). This proof is short — arithmetic on exponents — once its inputs are in place.

Remark 7.3 (Hardy assist, if needed). If Hypothesis 6.1 cannot be discharged from local jet-Gram geometry alone for the two-point pairing, the source-energy bound of Hypothesis 12.3 in §12 provides the missing room. Theorem 7.1 is then conditional on §12 as well as on the local inputs; the exponent inequality $A_{\text{toy}} < B_{\zeta}$ remains the only thing to verify.

8. FOUR ZEROS IN A SYMMETRIC QUARTET: STRONGER RIGIDITY

The full symmetry packet of an off-line zero is the quartet

$$\mathcal{Z}(\rho) = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}.$$

The four-center jet-Gram block carries strictly more rigidity than the two-point version: the toy visibility lower bound holds in a refined quotient that two-point comparisons cannot see, and the matching suppression hypothesis is correspondingly stronger. This section states the four-point closure independently of §7; either route suffices for §10.

Theorem 8.1 (Four-point closure). *Let ρ be a hypothetical off-line zero and $\mathcal{Z}(\rho)$ its symmetry quartet. The four-center jet-Gram comparison, under the same projection Π extended to the four-point quotient, yields $A_{\text{toy}}^{(4)} < B_{\zeta}^{(4)}$ for absolute four-point exponents, and hence $\mathcal{Z}(\rho)$ cannot be a quartet of zeros.*

Gap 8.2. State and prove the four-center analogue of $\hat{\Omega}_{\zeta}$, the four-point projection $\Pi_{(4)}$, and the four-point versions of Theorem 5.5 and Hypothesis 6.1; assemble the exponent inequality. v1 prose to mine: the local projective rigidity of the one-pair and multi-pair families, plus the corrected fixed-target quotient package.

Remark 8.3 (Hardy assist, if needed). The four-point analogue of Remark 7.3 applies: if the four-point standalone suppression cannot be proved purely locally, the source-energy bound of Hypothesis 12.3 can be invoked, with the same effect of supplying the gap between $A_{\text{toy}}^{(4)}$ and $B_{\zeta}^{(4)}$.

9. ODD-MOMENT CANCELLATION ACROSS N POINTS

The exponent inequality in Theorems 7.1 and 8.1 is the load-bearing inequality, but its inputs include a finite remainder coming from the Taylor expansion of Φ around midpoints. By symmetry, odd moments of the relevant expansion vanish under integration against the symmetric packet weight; this is the mechanism that keeps the remainders below $\min(Q^{-A_{\text{toy}}}, Q^{-B_{\zeta}})$ and lets the comparison close.

Lemma 9.1 (Odd-moment cancellation). *Let $\widehat{\mu}_{N,Q}$ be the symmetric N -point packet measure attached to height T and scale Q . For every $r \geq 0$,*

$$\partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0.$$

The same identity holds with the packet measure replaced by its convolution with any even bump $\widehat{\phi}$: for every $r \geq 0$,

$$\partial_z^{2r+1} (\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z))|_{z=0} = 0.$$

Gap 9.2. Pure parity argument; v1 has the full proof in the N -point odd-moment cancellation section. Migration is straight — only the symmetry packet definition needs to match Definition 2.1.

10. FROM LOCAL OBSTRUCTION TO A ZERO-FREE STRIP

The two-point and four-point closures of §7 and §8 exclude any single off-line zero above height T_0 ; to upgrade this to a zero-free strip we aggregate over heights using the polynomial nature of every loss in Lemmas 4.3, 3.5, Theorem 5.5, and Hypothesis 6.1.

Theorem 10.1 (Zero-free strip). *There is $T_0 > 0$ such that $\zeta(\beta + i\gamma) \neq 0$ for every $\beta \neq \frac{1}{2}$ and $|\gamma| \geq T_0$.*

Gap 10.2. Aggregate Theorem 7.1 or Theorem 8.1 across the dyadic height ladder; the polynomial losses combine into a single polynomial loss in $Q(\gamma)$ and the exponent inequality $A_{\text{toy}} < B_\zeta$ carries through uniformly.

11. THE RIEMANN HYPOTHESIS

Theorem 10.1 excludes high off-line zeros; standard verification of ζ on $|\gamma| \leq T_0$ covers the low-height range.

Theorem 11.1 (Riemann Hypothesis). *Every nontrivial zero of ζ satisfies $\text{Re } s = \frac{1}{2}$.*

Proof. Combine Theorem 10.1 with the standard low-height numerical verification of RH up to height T_0 . □ □

12. (OPTIONAL) EXTRA SOURCE ENERGY FROM ζ : THE HARDY RESERVOIR

This section is a self-contained interface. Its only role is to provide a single source-energy lower bound that Theorems 7.1 and 8.1 may invoke if their standalone forms in §7 / §8 prove insufficient. Nothing earlier in the paper depends on this section.

12.1. Canonical source curvature.

Definition 12.1 (Canonical source curvature). The canonical source curvature of ζ on the local chart of Definition 2.1 is

$$q''_{\text{can}} := \partial_t^3 Z_H(t),$$

where Z_H is the Hardy-normalized real critical-line scalar of ζ in the same chart.

Gap 12.2. Identification of q''_{can} with $\partial_t^3 Z_H$ is convention; the chart-regularity and zero-separation hypotheses of Definition 2.1 carry over verbatim. v1 prose to mine: the upstream Gate-1 arithmetic inputs section.

12.2. Source-energy reservoir.

Hypothesis 12.3 (Hardy source-energy bound). On retained packet intervals $I \subset I_T$, the canonical source curvature satisfies

$$\int_I |q''_{\text{can}}(m)|^2 dm \geq Q^{-A_E} |I|$$

on a positive polynomial-weight subfamily, for an absolute constant A_E .

Gap 12.4. Hardy-type lower bound on $\partial_t^3 Z_H$ across the retained packet family. This is an analytic estimate on ζ and not on the local jet-Gram geometry; it is what makes the section optional. v1 prose to mine: the Hardy-curvature mass-good packet source block.

Lemma 12.5 (Energy-to-suppression interface). *Hypothesis 12.3 implies Hypothesis 6.1 with absolute constant $B_\zeta = B_\zeta(A_E)$ satisfying $B_\zeta > A_{\text{toy}}$, via the carrier-corrected transfer of the source energy into the transverse channel of $\hat{\Omega}_\zeta(s; m)$.*

Gap 12.6. Carrier decomposition and packet trace step that takes $\int |q''_{\text{can}}|^2$ on the source side into a lower bound on $\Pi_{\hat{\Omega}_\zeta}$ on the jet-Gram side. v1 prose to mine: the canonical source-graded closure and carrier-corrected source bridge sections.

APPENDIX A. ARCHIVE MAP

The previous large draft is preserved in place at `rh/proof_of_rh.tex`; it is treated as a quarry from which prose and computations may be migrated, after restatement and refereeing, into the present file. Pointers to relevant v1 sections are given inline at each `wip` mark above; the table below summarizes the main mining correspondences.

Rebuild section	v1 source (in <code>proof_of_rh.tex</code>)
§2 (microscope)	Local kernel and jet normalization.
§3 (jet-Gram limits)	Finite- s local blocks; jet-limit local blocks.
§4 (finite blocks, whitening)	Corrected finite- s holomorphic whitening.
§5 (toy visibility)	Toy anomaly and transverse obstruction; leading value matrix and canonical calibration.
§6 (zeta suppression)	Odd holomorphic transverse channel; corrected fixed-target quotient package.
§7, §8 (two/four-point)	Local projective rigidity of the one-pair and multi-pair families; finite coefficient table contracts.
§9 (odd cancellation)	N -point odd-moment cancellation.
§10, §11 (strip, RH)	Master conditional theorem, introduction.
§12 (Hardy reservoir)	Hardy-curvature mass-good packet source block; carrier-corrected source bridge package; upstream Gate-1 arithmetic inputs.

The v1 sections labeled “Endgame architecture and remaining propagation problem,” “Downstream local closure package,” “Unified finite-survival architecture,” “Current proof state after the local package audits,” “How the older framing maps to the new framing,” and “No-go and audit ledger” are deliberately not migrated; their content is route history and process bookkeeping, not active proof material.